

PRACTICE SHEET 1

1. If 'n' be any natural number, then by which largest number ($n^3 - n$) is always divisible?
 (1) 3 (2) 6
 (3) 12 (4) 18
2. If n is an integer, then ($n^2 - n$) is always divisible by :
 (1) 4 (2) 5
 (3) 2 (4) 7
3. If the sum of the digits of any integer lying between 100 and 1000 is subtracted from the number, the result always is
 (1) Divisible by 6 (2) divisible by 2
 (3) Divisible by 9 (4) divisible by 5
4. If a number is divisible by both 11 and 13, then it must be necessarily:
 (1) Divisible by (11 + 13)
 (2) Divisible by (13 - 11)
 (3) Divisible by (11 × 13)
 (4) 429
5. If * is a digit such that 5824* is divisible by 11, then * equals:
 (1) 2 (2) 3
 (3) 5 (4) 6
6. If 78*3945 is divisible by 11, where * is a digit, then * is equal to
 (1) 1 (2) 0
 (3) 3 (4) 5
7. If the number 4 8 3 2 7 * 8 is divisible by 11, then the missing digit (*) is
 (1) 5 (2) 3
 (3) 2 (4) 1
8. Both the end digits of a 99 digit number N are 2. N is divisible by 11, then all the middle digits are:
 (1) 1 (2) 2
 (3) 3 (4) 4
9. If n is a whole number greater than 1, then $n^2(n^2 - 1)$ is always divisible by :
 (1) 16 (2) 12
 (3) 10 (4) 8
10. A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232, etc. Any number of this form is always exactly divisible by :
 (1) 7
 (2) 11
 (3) 13
 (4) Smallest 3-digit prime number
11. What least number, of 5 digits is divisible by 41?
 (1) 10045 (2) 10004
 (3) 10041 (4) 41000
12. The digit in unit's place of the product $81 \times 82 \times 83 \times \dots \times 89$ is
 (1) 0 (2) 2
 (3) 6 (4) 8
13. The digit in unit's place of the product $(2153)^{167}$ is:
 (1) 1 (2) 3
 (3) 7 (4) 9
14. The digit in the unit's place of the product $(2464)^{1793} \times (615)^{317} \times (131)^{491}$ is
 (1) 0 (2) 2
 (3) 3 (4) 5
15. Unit digit in $(264)^{102} + (264)^{103}$ is:
 (1) 0 (2) 4
 (3) 6 (4) 8
16. The digit in the unit's place of $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - 16^4 + 259]$ is:
 (1) 1 (2) 4
 (3) 5 (4) 6
17. The last digit of 3^{40} is
 (1) 1 (2) 3
 (3) 7 (4) 9
18. What will be the unit digit in the product 7^{105} ?
 (1) 5 (2) 7
 (3) 9 (4) 1
19. The unit digit in the expansion of $(2137)^{754}$ is
 (1) 1 (2) 3
 (3) 7 (4) 9
20. One's digit of the number $(22)^{23}$ is
 (1) 4 (2) 6
 (3) 8 (4) 2

- 21.** The unit digit in the product $(122)^{173}$ is
 (1) 2 (2) 4
 (3) 6 (4) 8
- 22.** Find the unit digit of $(7^{95} - 3^{58})$
 (1) 7 (2) 3
 (3) 4 (4) 0
- 23.** Find the unit digit of $111!$
 (1) 0 (2) 1
 (3) 5 (4) 3
- 24.** Unit digit in $(264)^{102} + (264)^{103}$ is:
 (1) 0 (2) 4
 (3) 6 (4) 8
- 25.** Find the unit digit in the product: $(4387)^{245} \times (621)^{72}$.
 (1) 1 (2) 2
 (3) 5 (4) 7
- 26.** The unit's digit in the product $7^{71} \times 6^{63} \times 3^{65}$ is
 (1) 1 (2) 2
 (3) 3 (4) 4
- 27.** Find the unit digit of the product of all the prime number between 1 and 99999
 (1) 9 (2) 7
 (3) 0 (4) N.O.T.
- 28.** Find the unit digit of $1^1 + 2^2 + 3^3 + \dots + 10^{10}$.
 (1) 9 (2) 7
 (3) 0 (4) N.O.T.
- 29.** Find the unit digit of
 $13^{24} \times 68^{57} + 24^{13} \times 57^{68} + 1234 + 5678$.
 (1) 4 (2) 7
 (3) 0 (4) 8
- 30.** The last digit of the following expression is:
 $(1!)^1 + (2!)^2 + (3!)^3 + (4!)^4 + \dots + (10!)^{10}$
 (1) 4 (2) 5
 (3) 6 (4) 7

ANSWER KEY

1.	(2)	2.	(3)	3.	(3)	4.	(3)	5.	(3)	6.	(4)	7.	(4)	8.	(4)	9.	(2)	10.	(4)
11.	(2)	12.	(1)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(1)	18.	(2)	19.	(4)	20.	(3)
21.	(1)	22.	(3)	23.	(1)	24.	(1)	25.	(4)	26.	(4)	27.	(3)	28.	(2)	29.	(1)	30.	(2)

PRACTICE SHEET 2

1. Find the number of zeroes at the end of the product 1000!
(1) 200 (2) 249
(3) 248 (4) 250
2. Find the number of zeroes at the end of the product 2700!
(1) 673 (2) 670
(3) 669 (4) 675
3. Find the number of zeroes at the end of the product $10 \times 15 \times 44 \times 28 \times 70$
(1) 2 (2) 3
(3) 4 (4) 5
4. Find the number of zeroes at the end of the product $2 \times 4 \times 6 \times \dots \times 48 \times 50$
(1) 6 (2) 12
(3) 7 (4) 5
5. Find the number of zeroes at the end of the product $1 \times 3 \times 5 \times 7 \times 9 \times 11 \times \dots \times 99 \times 101$
(1) 24 (2) 5
(3) 2 (4) 0
6. Find the number of zeroes at the end of the product $35 \times 36 \times 37 \times \dots \times 89 \times 90$
(1) 21 (2) 7
(3) 14 (4) 20
7. Find the number of zeroes at the end of the product $41 \times 42 \times \dots \times 109 \times 110$
(1) 26 (2) 9
(3) 17 (4) 25
8. Find the number of zeroes at the end of the product $25! \times 32! \times 45!$
(1) 10 (2) 23
(3) 22 (4) 7
9. Find the number of zeroes at the end of the product $1140! \times 358! \times 171!$
(1) 282 (2) 325
(3) 411 (4) 370
10. Find the number of zeroes at the end of the product $1^{1!} \times 2^{2!} \times 3^{3!} \times 4^{4!} \times \dots \times 10^{10!}$
(1) 51 (2) 10
(3) $5! + 10!$ (4) N.O.T
11. Find the number of Factors of 1728
(1) 28 (2) 29
(3) 30 (4) 31
12. Find the Number of Factor of 1420
(1) 12 (2) 13
(3) 14 (4) 15
13. Find the Number of Divisors of 10800
(1) 30 (2) 60
(3) 120 (4) 180
14. Find the No. of Prime Factor of 240.
(1) 4 (2) 5
(3) 6 (4) 8
15. Find the No. of prime factor $(30)^{26} \times (25)^{51} \times (12)^{23}$
(1) 249 (2) 250
(3) 255 (4) 260
16. Find the No. of Prime Factor $(30)^{15} \times (22)^{11} \times (15)^{24}$
(1) 110 (2) 115
(3) 120 (4) 125
17. Find the No. of prime factor of $(56)^{20} \times (36)^{31} \times (42)^{13} \times (13)^{21}$
(1) 240 (2) 242
(3) 264 (4) 248
18. Find the prime Factors 210
(1) 3 (2) 4
(3) 5 (4) 6
19. Find the sum of odd factors of 544
(1) 16 (2) 18
(3) 20 (4) 22
20. For the Number 2450 find
(i) Number of all factors
(ii) Number of even factors
(iii) Number of odd factors
(1) 18,9,9 (2) 18,10,8
(3) 18,8,10 (4) 18,12,6
21. When a number is divided by 893, the remainder is 193. What will be remainder when it is divided by 47?
(1) 3 (2) 5
(3) 25 (4) 33
22. $8.\overline{31} + 0.\overline{6} + 0.00\overline{2}$ is equal to
(1) $8.\overline{912}$ (2) $8.9\overline{12}$
(3) $8.9\overline{79}$ (4) $8.9\overline{79}$

- 23.** The value of $0.\overline{63} + 0.\overline{37}$ is
 (1) 1 (2) $100/99$
 (3) $99/100$ (4) $100/33$
- 24.** $(0.\overline{11} + 0.\overline{22}) \times 3$ is equal to
 (1) 3 (2) $1.\overline{9}$
 (3) 1 (4) $0.\overline{3}$
- 25.** $(1.\overline{2} \times 0.\overline{03}) =$
 (1) $0.\overline{04}$ (2) $0.\overline{036}$
 (3) $1.\overline{13}$ (4) $0.\overline{037}$
- 26.** The difference between $5.\overline{76}$ and $2.\overline{3}$ is
 (1) $2.\overline{54}$ (2) $3.\overline{73}$
 (3) $3.\overline{46}$ (4) $3.\overline{43}$
- 27.** The value of $0.\overline{2} + 0.\overline{3} + 0.\overline{32}$ is
 (1) $0.\overline{87}$ (2) $0.\overline{77}$
 (3) $0.\overline{82}$ (4) $0.\overline{86}$
- 28.** A number when divided by 280 leaves 115 as remainder. When the same number is divided by 35, the remainder is
 (1) 15 (2) 10
 (3) 20 (4) 17
- 29.** A certain number when divided by 175 leaves a remainder 132. When the same number is divided by 25, the remainder is:
 (1) 6 (2) 7
 (3) 8 (4) 9
- 30.** A positive integer when divided by 425 gives a remainder 45. When the same number is divided by 17, the remainder will be
 (1) 11 (2) 8
 (3) 9 (4) 10

ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(1)	5.	(4)	6.	(3)	7.	(3)	8.	(2)	9.	(3)	10.	(3)
11.	(1)	12.	(1)	13.	(2)	14.	(3)	15.	(1)	16.	(2)	17.	(3)	18.	(2)	19.	(2)	20.	(1)
21.	(2)	22.	(3)	23.	(2)	24.	(3)	25.	(4)	26.	(4)	27.	(1)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 3

1. Find the Remainder when $77 \times 85 \times 73$ is divided by 9
 (1) 1 (2) 2
 (3) 4 (4) 7
2. Find the Remainder when $273 + 375 + 478 + 657 + 597$ is divided by 25
 (1) 5 (2) 10
 (3) 9 (4) 8
3. Find the Remainder when $1330 \times 1356 \times 1363 \times 1368 \times 1397$ is divided by 13
 (1) 7 (2) 9
 (3) 11 (4) 8
4. Find the Remainder when $2327 + 2372 + 2394 + 4624 + 4650$ is divided by 23
 (1) 12 (2) 14
 (3) 13 (4) 10
5. Find the Remainder when 67^{32} is divided by 68
 (1) 67 (2) 66
 (3) 1 (4) 0
6. Find the Remainder when 99^{99} is divided by 100
 (1) 99 (2) 98
 (3) 1 (4) 3
7. Find the Remainder 197^{130} is divided by 196
 (1) 1 (2) 195
 (3) 7 (4) 5
8. Find the Remainder 6^{36} is divided by 215
 (1) 214 (2) 6
 (3) 5 (4) 1
9. Find the Remainder 75^{75} is divided by 37
 (1) 1 (2) 36
 (3) 3 (4) 7
10. Find the Remainder 43^{197} is divided by 7
 (1) 42 (2) 41
 (3) 1 (d) 6
11. Find the Remainder when $(59)^{28}$ is divided by 7
 (1) 2 (2) 4
 (3) 6 (4) 1
12. Find the Remainder when 41^{77} is divided by 17
 (1) 2 (2) 1
 (3) 6 (4) 4
13. Find the Remainder when 2^{49} is divided by 7
 (1) 1 (2) 2
 (3) 3 (4) 4
14. Find the Remainder when $(51^{203} + 2^{49})$ is divided by 17
 (1) 4 (2) 5
 (3) 6 (4) None of these
15. Find the Remainder when 8^{77} is divided by 7
 (1) 8 (2) 1
 (3) 13 (4) 7
16. Find the Remainder when $1! + 2! + 3! + 4! + \dots + 100!$ is divided by 5 is
 (1) 0 (2) 1
 (3) 2 (4) 3
17. Find the Remainder when $1! + 2! + 3! + 4! + \dots + 100!$ is divided by 6 is
 (1) 3 (2) 4
 (3) 2 (4) 1
18. Find the Remainder when 9^{111} is divided by 11:
 (1) 2 (2) 9
 (3) 7 (4) 6
19. Find the Remainder when 5^{2450} is divided by 126
 (1) 5 (2) 25
 (3) 125 (4) 1
20. Find the Remainder when 40^{1012} is divided by 7
 (1) 5 (2) 4
 (3) 3 (4) 2
21. Find the Remainder when $10^1 + 10 + 10^3 + \dots + 10^{100}$ is divided by 6
 (1) 4 (2) 6
 (3) 2 (4) 3
22. Find the Remainder when $10^1 + 10^2 + 10^3 + \dots + 10^{1000} + 10^{1001}$ is divided by 6
 (1) 4 (2) 6
 (3) 2 (4) 3

- 23.** Find the Remainder when 666666
134 times is divided by 13
(1) 1 (2) 3
(3) 11 (4) 9
- 24.** Find the Remainder when 555555
244 times is divided by 37
(1) 18 (2) 5
(3) 36 (4) 0
- 25.** Find the Remainder when 777777
363 times is divided by 11
(1) 0 (2) 7
(3) 1 (4) 3
- 26.** Find the Remainder when 888888 184
times is divided by 37
(1) 1 (2) 8
(3) 36 (4) 7
- 27.** Find the Remainder when 999999999 is
divided by 13
(1) 8 (2) 11
(3) 5 (4) 12
- 28.** Find the Remainder when 7^{99} is divided by
2400
(1) 1 (2) 49
(3) 343 (4) 7
- 29.** Find the Remainder when 3^{1989} is divided by
7
(1) 2 (2) 6
(3) 4 (4) 5
- 30.** Find the Remainder when 54^{124} is divided by
17
(1) 4 (2) 5
(3) 3 (4) 15

ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(2)	5.	(3)	6.	(1)	7.	(1)	8.	(4)	9.	(1)	10.	(3)
11.	(2)	12.	(3)	13.	(2)	14.	(4)	15.	(1)	16.	(4)	17.	(1)	18.	(2)	19.	(2)	20.	(4)
21.	(1)	22.	(3)	23.	(1)	24.	(2)	25.	(2)	26.	(2)	27.	(2)	28.	(3)	29.	(2)	30.	(1)

PRACTICE SHEET 4

1. $(7^{19} + 2)$ is divided by 6, the remainder is :
 (1) 5 (2) 3
 (3) 2 (4) 1
2. If two numbers are each divided by the same divisor, the remainders are respectively 3 and 4. If the sum of the two numbers be divided by the same divisor, the remainder is 2. The divisor is
 (1) 9 (2) 7
 (3) 5 (4) 3
3. The remainder when 3^{21} is divided by 5 is
 (1) 1 (2) 2
 (3) 3 (4) 4
4. $9^6 - 11$ when divided by 8 would leave a remainder of :
 (1) 0 (2) 1
 (3) 2 (4) 6
5. If 17^{200} is divided by 18, the remainder is—
 (1) 17 (2) 16
 (3) 1 (4) 2
6. When 2^{31} is divided by 5 the remainder is
 (1) 4 (2) 3
 (3) 2 (4) 1
7. $(4^{61} + 4^{62} + 4^{63})$ is divisible by
 (1) 3 (2) 11
 (3) 13 (4) 17
8. $(2^{71} + 2^{72} + 2^{73} + 2^{74})$ is divisible by
 (1) 9 (2) 10
 (3) 11 (4) 13
9. $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ is divisible by
 (1) 11 (2) 16
 (3) 25 (4) 30
10. $(49)^{15} - 1$ is exactly divisible by :
 (1) 50 (2) 51
 (3) 29 (4) 8
11. The sum of three consecutive odd natural numbers is 147. Then, the middle number is:
 (1) 47 (2) 48
 (3) 49 (4) 51
12. The sum of first 20 odd natural numbers is equal to :
 (1) 210 (2) 300
 (3) 400 (4) 420
13. The sum of all natural numbers from 75 to 97 is :
 (1) 1598 (2) 1798
 (3) 1958 (4) 1978
14. The sum of all natural numbers between 100 and 200, which are multiples of 3 is :
 (1) 5000 (2) 4950
 (3) 4980 (4) 4900
15. The sum of the squares of three consecutive natural numbers is 2030. Then, what is the middle number?
 (1) 25 (2) 26
 (3) 27 (4) 28
16. The sum of three consecutive odd natural numbers is 87. The smallest of these numbers is :
 (1) 29 (2) 31
 (3) 23 (4) 27
17. Sum of three consecutive even integers is 54. Find the least among them.
 (1) 18 (2) 15
 (3) 14 (4) 16
18. The sum of three consecutive numbers is 87. The middle number is
 (1) 27 (2) 29
 (3) 30 (4) 28
19. What is the sum of two consecutive even numbers, the difference of whose square is 84?
 (1) 38 (2) 34
 (3) 42 (4) 46
20. The sum of all the natural numbers from 51 to 100 is
 (1) 5050 (2) 4275
 (3) 4025 (4) 3775
21. A number when divided successively by 4 and 5 leaves remainder 1 and 4 respectively.

When it is successively divided by 5 and 4 the respective remainders will be

- (1) 4, 1 (2) 3, 2
(3) 2, 3 (4) 1, 2

22. The expression $2^{6n} - 4^{2n}$, where n is a natural number is always divisible by

- (1) 15 (2) 18
(3) 36 (4) 48

23. When an integer K is divided by 3, the remainder is 1, and when $K + 1$ is divided by 5, the remainder is 0. Of the following, a possible value of K is

- (1) 62 (2) 63
(3) 64 (4) 65

24. Which one of the numbers is divisible by 25?

- (1) 303310 (2) 373355
(3) 303375 (4) 22040

25. For any integral value of n , $3^{2n} + 9n + 5$ when divided by 3 will leave the remainder

- (1) 1 (2) 2
(3) 0 (4) 5

26. The maximum value of F in the following equation $5E9 + 2F8 + 3G7 = 1114$ is where E, F, G each stands for any digit.

- (1) 8 (2) 9
(3) 7 (4) 5

27. The number $334 \times 545 \times 7p$ is divisible by 3340 if p is at least.

- (1) 2 (2) 4
(3) 3 (4) 1

28. The sum of three numbers is 252. If the first number is thrice the second and third number is two-third of the first, then the second number is

- (1) 41 (2) 21
(3) 42 (4) 84

29. The difference between the greatest and the least five-digit numbers formed by the digits 2, 5, 0, 6 and 8 is (repetition of digits is not allowed)

- (1) 69552 (2) 65925
(3) 65952 (4) 63952

30. A man has some hens and some cows. If the total number of heads of hens and cows together is 50 and the number of feet of hens and cows together is 142, then the number of cows is

- (1) 21 (2) 25
(3) 27 (4) 29

ANSWER KEY

1.	(2)	2.	(3)	3.	(3)	4.	(4)	5.	(3)	6.	(2)	7.	(1)	8.	(2)	9.	(4)	10.	(4)
11.	(3)	12.	(3)	13.	(4)	14.	(2)	15.	(2)	16.	(4)	17.	(4)	18.	(2)	19.	(3)	20.	(4)
21.	(3)	22.	(4)	23.	(3)	24.	(3)	25.	(2)	26.	(4)	27.	(1)	28.	(3)	29.	(3)	30.	(1)

PRACTICE SHEET 5

1. A number divided by 13 leaves a remainder 1 and if the quotient, thus obtained, is divided by 5, we get a remainder of 3. What will be the remainder if the number is divided by 65 ?
 (1) 28 (2) 16
 (3) 18 (4) 40
2. If two numbers are each divided by the same divisor, the remainders are respectively 3 and 4. If the sum of the two numbers be divided by the same divisor, the remainder is 2. The divisor is
 (1) 9 (2) 7
 (3) 5 (4) 3
3. A number consists of two digits. If the number formed by interchanging the digits is added to the original number, the resulting number (i.e. the sum) must be divisible by
 (1) 11 (2) 9
 (3) 5 (4) 3
4. What is the value of x for which $x, x + 1, x + 3$ are all prime numbers?
 (1) 0 (2) 1
 (3) 2 (4) 101
5. Which one of the following is correct regarding the number 222222 ?
 (1) It is divisible by 3 but not divisible by 7
 (2) It is divisible by 3 and 7 but not divisible by 11
 (3) It is divisible by 2 and 7 but not divisible by 11
 (4) It is divisible by 3, 7 and 11
6. What is the sum of positive integers less than 100 which leave remainder 1 when divided by 3 and leave a remainder 2 when divided by 4?
 (1) 416 (2) 620
 (3) 1250 (4) 1314
7. A three-digit number is divisible by 11 and has its digit in the unit's place equal to 1. The number is 297 more than the number obtained by reversing the digits. What is the number?
 (1) 121 (2) 231
 (3) 561 (4) 451
8. The remainder on dividing given integers a and b by 7 are, 5 and 4 respectively. What is the remainder when ab is divided by 7?
 (1) 3 (2) 4
 (3) 5 (4) 6
9. Which one of the following is the largest 4 digit number divisible by 88?
 (1) 9988 (2) 9966
 (3) 9944 (4) 8888
10. The number 2784936 is divisible by which of the following number?
 (1) 86 (2) 87
 (3) 88 (4) 89
11. If k is any even positive integer, then $(k^2 + 2k)$ is
 (1) divisible by 24
 (2) divisible by 8 but may not be divisible by 24
 (3) divisible by 4 but may not be divisible by 8
 (4) divisible by 2 but may not be divisible by 4
12. What is the number of total three digit numbers with unit digit 7 and divisible by 11 ?
 (1) 6 (2) 7
 (3) 8 (4) 9
13. In a two digit number if it is known that its units digit exceeds its tens digit by 2 and that the product of the given number and the sum of its digits is equal to 144, then the number is
 (1) 46 (2) 42
 (3) 26 (4) 24
14. The largest integer that divides product of any four consecutive integers is
 (1) 4 (2) 6
 (3) 12 (4) 24
15. A ten digit number is divisible by 4 as well as by 5. What could be the possible digit at ten's place in the given number?
 (1) 0, 1, 2, 4 and 6 (2) 1, 2, 4, and 8
 (3) 2, 3, 4, 6 and 8 (4) 0, 2, 4, 6 and 8

16. If $\frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

- (a) 1
- (b) 2
- (c) 3
- (d) Cannot be determined due to insufficient data

17. Consider the following statements :

If p is a prime number such that $p + 2$ is also a prime then

I. $p(p + 2) + 1$ is a perfect square.

II. 12 is a divisor of $p + (p + 2)$, if $p > 3$

Which of the statements given above is/are correct

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

18. A number consists of two digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of digits in the number is :

- (1) 8
- (2) 9
- (3) 6
- (4) 7

19. Of the three numbers, the second is twice the first and it is also thrice the third. If the average of three numbers is 44, the difference of the first number and the third number is :

- (1) 24
- (2) 18
- (3) 12
- (4) 6

20. A two digit number is five times the sum of its digits. If 9 is added to the number, the digits interchange their positions. The sum of digits of the number is :

- (1) 11
- (2) 9
- (3) 7
- (4) 6

21. How many numbers less than 1000 are multiples of both 10 and 13?

- (1) 9
- (2) 8
- (3) 6
- (4) 7

22. The number 1, 2, 3, 4, ..., 1000 are multiplied together. The number of zeros at the end (on the right) of the product must be:

- (1) 30
- (2) 200
- (3) 211
- (4) 249

23. Consider the following assumption and two statements:

Assumption: A number, 'ABCDE' is divisible by 11.

Statement I: $E - D + C - B + A$ is divisible by 11.

Statement II: $E - D + C - B + A = 0$

Which one of the following is correct ?

- (a) Only statement I can be drawn from the assumption
- (b) Only statement II can be drawn from the assumption
- (c) Both the statements can be drawn from the assumption
- (d) Neither of the statements can be drawn from the assumption

24. The product of two numbers is 0.008. One of the number is $\frac{1}{5}$ of the other. The smaller number is

- (1) 0.2
- (2) 0.4
- (3) 0.02
- (4) 0.04

25. I multiplied a natural number by 18 and another by 21 and added the products. Which one of the following could be the sum?

- (1) 2007
- (2) 2008
- (3) 2006
- (4) 2002

26. If the sum of two numbers be multiplied by each number separately, the products so obtained are 247 and 114. The sum of the numbers is

- (1) 19
- (2) 20
- (3) 21
- (4) 23

27. If a and b are odd numbers, then which of the following is even ?

- (1) $a + b + ab$
- (2) $a + b - 1$
- (3) $a + b + 1$
- (4) $a + b + 2ab$

28. If two numbers x and y separately divided by a number d, remainders obtained are 4375 and 2986 respectively. If the sum of the numbers i.e. $(x+y)$ is divided by the same number d remainder obtained is 2361. The value of number d is:

- (1) 7361
- (2) 5000
- (3) 4000
- (4) 2542

- 29.** A farmer divides his herd of n cows among his four sons so that the first son gets one – half the herd, the second son gets one – fourth, the third son gets one – fifth and the fourth son gets 7 cows. The value of n is
 (1) 80 (2) 100
 (3) 140 (4) 180

- 30.** In an examination, a student scores 4 marks for every correct answer and loses 1 mark

for every wrong answer. A student attempted all the 200 questions and scored in all 200 marks. The number of questions, he answered correctly was

- (1) 82 (2) 80
 (3) 68 (4) 60

ANSWER KEY

1.	(4)	2.	(3)	3.	(1)	4.	(3)	5.	(4)	6.	(1)	7.	(4)	8.	(4)	9.	(3)	10.	(3)
11.	(2)	12.	(3)	13.	(4)	14.	(4)	15.	(4)	16.	(3)	17.	(3)	18.	(3)	19.	(3)	20.	(2)
21.	(4)	22.	(4)	23.	(3)	24.	(4)	25.	(1)	26.	(1)	27.	(4)	28.	(2)	29.	(3)	30.	(2)

CDS PYQ

1. The two digit number, which when divided by sum of the digit and product of the digits, respectively. The remainder is same and the difference of quotients is one, is
[CDS 2013 (I)]
(a) 14 (b) 23
(c) 32 (d) 41
2. Which one of the following numbers is divisible by 11?
[CDS 2013 (1)]
(a) 45678940 (b) 54857266
(c) 87524398 (d) 93455120
3. If N , $(N + 2)$ and $(N + 4)$ are prime number, then the number of possible solution for N are:
[CDS 2013 (1)]
(a) 1 (b) 2
(c) 3 (d) None of these
4. The smallest positive prime (say p) such that $2^p - 1$ is not a prime is:
[CDS 2013 (1)]
(a) 5 (b) 11
(c) 17 (d) 29
5. Consider the following statements:
I. There are a finite number of rational numbers between any two rational numbers.
II. There is an infinite number of rational numbers between any two rational numbers
III. There are a finite number of irrational numbers between any two rational numbers
Which of the above statements is/are correct?
[CDS 2013 (1)]
(a) Only I (b) Only II
(c) Only III (d) Both I and II
6. Representation of $0.\overline{2341}$ in the form p/q , where p and q are integers, $q \neq 0$, is
[CDS]
(a) $781/3330$ (b) $1171/4995$
(c) $2341/9990$ (d) $2339/9990$
7. If m and n are natural number, then $\sqrt[n]{m}$ is:
[CDS 2013 (2)]
(a) Always irrational
(b) Irrational unless n is the math power of an integer
(c) Irrational unless m is the n th power of an integer
(d) Irrational unless m is the n are co prime
8. If x is positive even integer and y is negative odd integer, then x^y is:
[CDS 2013 (2)]
(a) Odd integer (b) Even integer
(c) Rational number (d) None of these
9. The paid of rational number that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is:
[CDS 2013 (2)]
(a) $\frac{262}{1000}, \frac{752}{1000}$ (b) $\frac{24}{100}, \frac{78}{100}$
(c) $\frac{9}{40}, \frac{31}{40}$ (d) $\frac{252}{1000}, \frac{748}{1000}$
10. What is the last digit in $7^{402} + 3^{402}$?
[CDS 2013 (2)]
(a) 0
(b) 4
(c) 8
(d) None of the above
11. What is $26^2 + 97^2$ equal to?
[CDS 2013 (2)]
(a) $27^2 + 93^2$ (b) $34^2 + 93^2$
(c) $82^2 + 41^2$ (d) $79^2 + 62^2$
12. Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ (where, k and m are integer), when:
[CDS 2013 (2)]
(a) k is odd
(b) k is even
(c) k can be both odd and even
(d) No such form is possible
13. If k is a positive integer, then every square integer is of the form:
[CDS 2013 (2)]
(a) Only $4k$ (b) $4k$ or $4k + 3$
(c) $4k + 1$ or $4k + 3$ (d) $4k$ or $4k + 1$
14. If b is the largest square divisor of c and a^2 divides c , then which one of the following is correct (where a , b and c are integers)?
[CDS 2013 (2)]

- (a) b divides a
(b) a does not divide b
(c) a divided b
(d) Neither 10 nor 20
- 15.** $19^5 + 21^5$ is divisible by:
[CDS 2013 (2)]
(a) Only 10
(b) Only 20
(c) Both 10 and 20
(d) Neither 10 nor 20
- 16.** Consider the following statements:
I. 7710312401 is divisible by 11
II. 173 is a primer number
Which of the statements given above is/are correct?
[CDS 2013 (2)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 17.** Consider the following statements:
I. To obtain prime number less than 121, we have to reject all the multiples of 2,3,5 and 7
II. Every composite number less than 121 is divisible by a prime number less than 11
Which of the statements given above is/are correct?
[CDS 2013 (2)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 18.** Consider the following statements:
I. No integer of the form $4k + 3$, where k is an integer can be expressed as the sum of two squares.
II. Square of an odd integer can express in the form $8k + 1$, where k is an integer.
Which of the statements given above is/are correct?
[CDS 2014 (1)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 19.** If n is a whole number greater than 1, then $n^2 (n^2 - 1)$ is always divisible by:
[CDS 2014 (1)]
(a) 12 (b) 24
(c) 48 (d) 60
- 20.** What is the remainder when 4^{1000} is divided by 7?
[CDS 2014 (1)]
(a) 1 (b) 2
(c) 4 (d) None of these
- 21.** The difference of two consecutive cubes:
[CDS 2014 (1)]
(a) Is odd or even
(b) Is never divisible by 2
(c) Is always even
(d) None of these
- 22.** The product of four consecutive natural numbers plus one is:
[CDS 2014 (1)]
(a) A non square
(b) Always sum of two square numbers
(c) A square
(d) None of the above
- 23.** $7^{10} - 5^{10}$ is divisible by:
[CDS 2014 (2)]
(a) 10 (b) 7
(c) 5 (d) 11
- 24.** The multiplication of a three digits number XY5, with digit Z yields X215. What is $X + Y + Z$ equal to?
[CDS 2014 (2)]
(a) 13 (b) 15
(c) 17 (d) 18
- 25.** If $N^2 - 33$, $N^2 - 31$ and $N^2 - 29$ are prime numbers, then what is the number of possible values of N, where N is integer?
[CDS 2014 (2)]
(a) 1 (b) 2
(c) 6 (d) None of these
- 26.** How many pairs of positive integers m and n satisfy the equation $\frac{1}{m} + \frac{4}{n} = \frac{1}{12}$, where n is an odd integers less than 60?
[CDS 2014 (2)]
(a) 7 (b) 5
(c) 4 (d) 3
- 27.** Consider all those two digits positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum?
[CDS 2014 (2)]
(a) 310 (b) 314
(c) 218 (d) 323
- 28.** How many pairs of X and Y are possible in the number 763X4Y2, if the number is divisible by 9?
[CDS 2014 (2)]

(a) 8
(c) 10

(b) 9
(d) 11

29. What is the remainder when 4^{1012} is divided by 7?

[CDS 2014 (2)]

(a) 1
(c) 3

(b) 2
(d) 4

30. What is the remainder when $(17^{23} + 23^{23} + 29^{23})$ is divided by 23?

[CDS 2014 (2)]

(a) 0
(c) 2

(b) 1
(d) 3

31. p, q and r are prime numbers such that $p < q < r < 13$. In how many cases would $(p + q + r)$ also be a prime number?

[CDS 2014 (2)]

(a) 1
(c) 3

(b) 2
(d) None of these

32. What is the number of divisors of 360?

[CDS 2014 (2)]

(a) 12
(c) 24

(b) 18
(d) None of these

33. If $a_n = 3 - 4n$, then what is $a_1 + a_2 + a_3 + \dots + a_n$ equal to?

$$\left[1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2} \right]$$

[CDS 2014 (2)]

(a) $-n(4n-3)$
(c) $-n^2$

(b) $-n(2n-1)$
(d) $-n(2n+1)$

34. What is the remainder when $(1235 \times 4523 \times 2451)$ is divided by 12?

[CDS 2014 (2)]

(a) 1
(c) 5

(b) 3
(d) 7

35. A certain number of two digits is three times the sum of its digits. If 45 is added to the number, the digits will be reversed. What is the sum of the squares of the two digits of the number?

[CDS 2014 (2)]

(a) 41
(c) 53

(b) 45
(d) 64

36. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

[CDS 2015 (1)]

(a) 1
(b) 2
(c) 3
(d) Cannot be determined due to insufficient data

37. The digit in the units place of the product $81 \times 82 \times 83 \times 84 \times \dots \times 99$ is:

[CDS 2015 (1)]

(a) 0
(c) 6

(b) 4
(d) 8

38. A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was:

[CDS 2015 (1)]

(a) 12
(c) 25

(b) 15
(d) 32

39. What is the number of possible pairs of (P, Q) if the number 357P25Q is divisible by both 3 and 5?

[CDS 2015 (1)]

(a) 7
(b) 6
(c) 5
(d) None of the above

40. The difference between the square of two consecutive odd integers is always divisible by:

[CDS 2015 (1)]

(a) 3
(c) 8

(b) 7
(d) 16

41. What is the maximum value of m if the number $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$ is divisible by 5^m ?

[CDS 2015 (1)]

(a) 4
(c) 6

(b) 5
(d) 7

42. When a ball bounces, it rises to $\frac{2}{3}$ of the height from which it fell. If the ball is dropped from a height of 36m, how high will it rise at the third bounce?

[CDS 2015 (1)]

(a) $10\frac{1}{3}m$

(b) $10\frac{2}{3}m$

(c) $12\frac{1}{3}m$

(d) $12\frac{2}{3}m$

- 43.** A light was seen regularly at an interval of 13 seconds. It was seen for the first time at 1 hour 54 minutes 50 seconds (am) and the last time at 3 hours 17 minutes 49 seconds (am). How many times was the light seen?

[CDS 2015 (1)]

(a) 375
(c) 283

(b) 378
(d) 384

- 44.** If n is natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is:

[CDS 2015 (1)]

(a) $x_1 + x_2 + x_3$

(b) $x_1 x_2 x_3$

(c) $(x_1 + 1)(x_2 + 1)(x_3 + 1)$

(d) None of the above

- 45.** Consider the following statements for the sequence of numbers given below:

11, 111, 1111, 11111

1. Each number can be expressed in the form $(4m + 3)$, where m is a natural number

2. Some numbers are squares.

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

- 46.** Consider the following statements:

1. There exists only one prime number p such that $(17p + 1)$ is a square

2. If x is the product of 10 consecutive prime numbers starting from 2, then $(x+1)$ is also prime number.

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

- 47.** The last digit in the expansion of 17^{256} is:

[CDS 2015 (1)]

(a) 9

(b) 7

(c) 3

(d) 1

- 48.** What is the remainder obtained when $1421 \times 1423 \times 1425$ is divided by 12?

[CDS 2015 (1)]

(a) 1
(c) 3

(b) 2
(d) 4

- 49.** What is the remainder when 4^{96} is divided by 6?

[CDS 2015 (1)]

(a) 4
(c) 2

(b) 3
(d) 1

- 50.** Let p be a prime number other than 2 or 5. One would like to express the vulgar fraction $1/p$ in the form of a recurring decimal. Then the decimal will be

[CDS 2015 (1)]

(a) a pure recurring decimal and its period will be necessarily $(p - 1)$

(b) a mixed recurring decimal and its period will be necessarily $(p - 1)$

(c) a pure recurring decimal and its period will be some factor of $(p - 1)$

(d) a mixed recurring decimal and its period will be some factor of $(p - 1)$

- 51.** The number of values of x satisfying $x + \frac{100}{x} > 50$, where x is a natural number less than or equal to 100 is:

[CDS 2015 (2)]

(a) 51

(b) 53

(c) 55

(d) 57

- 52.** The largest natural number which divides every natural number of the form $(n^3 - n)$ ($n - 2$), where n is a natural number greater than 2 is:

[CDS 2015 (2)]

(a) 6

(b) 12

(c) 24

(d) 48

- 53.** The digit in the units place of the resulting number of the expression $(234)^{100} + (234)^{101}$ is:

[CDS 2015 (2)]

(a) 6

(b) 4

(c) 2

(d) 0

- 54.** The seven digit number $876p37q$ is divisible by 225. The values of p and q can be respectively:

[CDS 2015 (2)]

(a) 9, 0

(b) 0, 0

(c) 0, 5

(d) 9, 5

- 55.** Let x and y be positive integers such that $x > y$. The expressions $3x + 2y$ and $2x + 3y$ when

divided by 5 leave remainder 2 and 3 respectively. What is the remainder when $(x-y)$ is divided by 5?

[CDS 2015 (2)]

- (a) 4 (b) 2
(c) 1 (d) 0

56. A number consists of two digits, whose sum is 7. If the digits are reversed, the number is increased by 27. The product of digits of the number is:

[CDS 2015 (2)]

- (a) 6 (b) 8
(c) 10 (d) 12

57. Consider all positive two digit numbers each of which when divided by 7 leaves a remainder 3. What is their sum?

[CDS 2015 (2)]

- (a) 661 (b) 666
(c) 676 (d) 777

58. The value of $0.\overline{63} + 0.\overline{37}$ is

[CDS 2015 (2)]

- (a) 1 (b) $100/91$
(c) $100/99$ (d) $1000/99$

59. Consider the following statements in respect of the expression

$$S_n = \frac{n(n+1)}{2}, \text{ where } n \text{ is an integer.}$$

1. There are exactly two values of n for which $S_n = 861$

2. $S_n = S_{-(n+1)}$ and hence for any integer m , we have two values of n for which $S_n = m$.

Which of the above statement is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) neither 1 nor 2

60. Consider the following statements in respect of two different non-zero integers' p and q .
I. For $(p+q)$ to be less than $(p-q)$, q must be negative.

II. For $(p+q)$ to be greater than $(p-q)$, both p and q must be positive.

Which of the above statements is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) neither 1 nor 2

61. $7^{10} - 5^{10}$ is divisible by:

[CDS 2016 (1)]

- (a) 5 (b) 7

- (c) 10 (d) none

62. Let a two digit number be k times the sum of its digits. If the number formed by interchanging the digit is m times the sum of the digits, then the value of m is:

[CDS 2016 (1)]

- (a) $9-k$ (b) $10-k$
(c) $11-k$ (d) $k-1$

63. Let S be a set of first fourteen natural numbers. The possible number of pairs (a, b) , where $a, b, \in S$ and $a \neq b$ such that ab leaves remainder 1 when divided by 15, is:

[CDS 2016 (1)]

- (a) 3
(b) 5
(c) 6
(d) None of the above

64. A clock strikes once 1 O'clock, twice at 2 O'clock and thrice at 3 O'clock and so on. If it takes 8 seconds to strike at 5 O'clock, the time taken by it to strike at 10 O'clock is:

[CDS 2016 (1)]

- (a) 14 seconds
(b) 16 seconds
(c) 18 seconds
(d) None of the above

65. What is the maximum value of m , if the number $N = 90 \times 42 \times 324 \times 55$ is divisible by 3^m ?

[CDS 2016 (1)]

- (a) 8 (b) 7
(c) 6 (d) 5

66. Consider the following statements:

1. Every natural number is a real number
 2. Every real number is a rational number
 3. Every integer is a real number
 4. Every rational number is a real number
- Which of the above statements are correct?

[CDS 2016 (1)]

- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2 and 3 only (d) 3 and 3 only

67. Consider the following statements:

1. There exists a positive real number m such that $\cos x = 2^{m+1}$.
2. $mn \geq m + n$ for all m, n belonging to set of natural numbers.

Which of the above statements is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only

- (c) Both 1 and 2 (d) neither 1 nor 2

68. If m and n are distinct natural numbers, then which of the following is/are integer/integers?

1. $\frac{m}{n} + \frac{n}{m}$
2. $mn \left(\frac{m}{n} + \frac{n}{m} \right) (m^2 + n^2)$
3. $\frac{mn}{m^2 + n^2}$

Select the correct answer using the code given below:

[CDS 2016 (1)]

- (a) 1 and 2 (b) 2 only
(c) 2 and 3 (d) 3 only

69. Outside a meeting room, Madhukar was told by a person that each meeting takes place after $13/4$ hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 pm. At what time did Madhukar receive this information?

[CDS 2016 (1)]

- (a) 10:20 am (b) 11:30 am
(c) 11:40 am (d) 11:50 am

70. If $\frac{61}{19} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

[CDS 2016 (1)]

- (a) 1
(b) 2
(c) 3
(d) Cannot be determined due to insufficient data

71. What would be the maximum value of Q in the equation $5P9 + 3R7 + 2Q8 = 1114$?

[CDS 2016 (2)]

- (a) 9 (b) 8
(c) 5 (d) 4

72. In an examination, a student was asked to divide a certain number by 8. By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was the number?

[CDS 2016 (2)]

- (a) 252 (b) 256
(c) 258 (d) 260

73. A boy saves Rs 4.65 daily. What is the least number of days in which he will be able to save an exact number of rupees?

[CDS 2016 (2)]

- (a) 10 (b) 20
(c) 21 (d) 25

74. What is the unit digit of 7^{139} ?

[CDS 2016 (2)]

- (a) 9 (b) 7
(c) 6 (d) 3

75. Which of the following rational number has non-terminating and repeating decimal expansion?

[CDS 2016 (2)]

- (a) $15/1600$ (b) $23/8$
(c) $35/50$ (d) $17/6$

76. The sum of squares of two positive integers is 208. If the square of the larger number is 18 times the smaller number, then what is the difference of the larger and smaller number?

[CDS 2016 (2)]

- (a) 2 (b) 3
(c) 4 (d) 6

77. What is the remainder when 2^{100} is divided by 101?

[CDS 2016 (2)]

- (a) 1 (b) 11
(c) 99 (d) 100

78. Which one of the following is correct in respect of the number 1729?

[CDS 2016 (2)]

- (a) It cannot be written as the sum of the cube of two positive integers
(b) It can be written as the sum of the cubes of two positive integers in one way only
(c) It can be written as the sum of the cubes of two positive integers in two ways only
(d) It can be written as the sum of the cubes of two positive integers in three ways only

79. Consider the following statements in respect of positive odd integers x and y :

1. $x^2 + y^2$ is even integer.
2. $x^2 + y^2$ is divisible by 4

Which of the above statements is/are correct?

[CDS 2016 (2)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 80.** What is the remainder when $13^5 + 14^5 + 15^5 + 16^5$ is divided by 29?
[CDS 2016 (2)]
(a) 8 (b) 5
(c) 3 (d) 0
- 81.** What is the difference between the sum of the cubes and that of squares of first ten natural numbers?
[CDS 2016 (2)]
(a) 2280 (b) 2640
(c) 3820 (d) 4130
- 82.** $2^{122} + 4^{62} + 8^{42} + 4^{64} + 2^{130}$ is divisible by which one of the following integers?
[CDS 2016 (2)]
(a) 3 (b) 5
(c) 7 (d) 11
- 83.** What is the remainder when the number $(4444)^{4444}$ is divided by 9?
[CDS 2017 (1)]
(a) 4 (b) 6
(c) 7 (d) 8
- 84.** The number of prime numbers which are less than 100 is:
[CDS 2017 (1)]
(a) 24 (b) 25
(c) 26 (d) 27
- 85.** Consider the following statements:
1. Of two consecutive integers, one is even
2. Square of an odd integer is of the form $8n+1$
Which of the above statements is/are correct?
[CDS 2017 (1)]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 86.** $(N^{p-1} - 1)$ is a multiple of p , if N is prime to p and p is a:
[CDS 2017 (1)]
(a) Prime number
(b) Rational number
(c) Real number
(d) Composite number
- 87.** Which one of the following is largest
[CDS 2017 (1)]
(a) $7/9$ (b) $11/14$
(c) $3/4$ (d) $10/13$
- 88.** How many numbers between 500 and 1000 are divisible by 13?
- [CDS 2017 (2)]**
(a) 36 (b) 37
(c) 38 (d) 39
- 89.** If the points P and Q represent the real numbers $0.8\bar{3}$ and $0.6\bar{2}$ on the number line, then the distance between P and Q is
[CDS 2017 (2)]
(a) $21/90$ (b) $19/90$
(c) $21/100$ (d) $56/90$
- 90.** Three numbers which are co-prime to each other, are such that the product of the first two is 286 and that of the last two is 770. What is the sum of the three numbers?
[CDS 2018 (1)]
(a) 85 (b) 80
(c) 75 (d) 70
- 91.** All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product?
[CDS 2018 (1)]
(a) 0
(b) 80
(c) 5
(d) None of the above
- 92.** Which one of the following is correct?
[CDS 2018 (1)]
(a) Decimal expansion of a rational number is terminating.
(b) Decimal expansion of a rational number is non-terminating.
(c) Decimal expansion of an irrational number is terminating.
(d) Decimal expansion of an irrational number is non-terminating and non-repeating.
- 93.** What is the difference between $0.\bar{9}$ and 0.9?
[CDS 2018 (1)]
(a) 0 (b) 0.099
(c) 0.1 (d) 0.09
- 94.** Which one of the following decimal number is a rational number with denominator 37?
[CDS 2018 (1)]
(a) 0.459459459... (b) 0.459459459
(c) 0.045949459... (d) 0.00459459...
- 95.** What is the largest power of 10 that divides the product $1 \times 2 \times 3 \times 4 \times \dots \times 23 \times 24 \times 25$?
[CDS 2018 (1)]

- (a) 2
(b) 4
(c) 5
(d) None of the above

96. $5^{17} + 5^{18} + 5^{19} + 5^{20}$ is divisible by:

[CDS 2018 (1)]

- (a) 7
(b) 9
(c) 11
(d) 13

97. How many five digit numbers of the form $XXYXX$ is/are divisible by 33?

[CDS 2018 (2)]

- (a) 1
(b) 3
(c) 5
(d) Infinite

98. A five digit number $XY235$ is divisible by 3 where X and Y are digits satisfying $X + Y \leq 5$. What is the number of possible pairs of values of (X, Y) ?

[CDS 2018 (2)]

- (a) 5
(b) 6
(c) 7
(d) 9

99. The number of divisors of the number 38808, exclusive of the divisors 1 and itself, is:

[CDS 2018 (2)]

- (a) 74
(b) 72
(c) 70
(d) 68

100. If α and β are two real numbers such that $\alpha + \beta = -\frac{q}{p}$ and $\alpha\beta = \frac{r}{p}$, where $1 < p < q < r$, then which one of the following is the greatest?

[CDS 2018 (2)]

- (a) $\frac{1}{\alpha + \beta}$
(b) $\frac{1}{\alpha} + \frac{1}{\beta}$
(c) $-\frac{1}{\alpha\beta}$
(d) $\frac{\alpha\beta}{\alpha + \beta}$

101. Consider the following statements in respect of three 3-digit numbers XYZ , YZX and ZXY :

1. The sum of the numbers is not divisible by $(X+Y+Z)$.

2. The sum of the numbers is divisible by 111.

Which of the above statements is/are correct?

[CDS 2018 (2)]

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

102. The number of all pairs (m, n) , where m and n are positive integers, such that

$$\frac{1}{m} + \frac{1}{n} - \frac{1}{mn} = \frac{2}{5} \text{ is:}$$

[CDS 2018 (2)]

- (a) 6
(b) 5
(c) 4
(d) 2

103. The sum of all possible product taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4$ is:

[CDS 2018 (2)]

- (a) 0
(b) -30
(c) 30
(d) 55

104. If a, b and c are positive integers such that

$$\frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{2}}}} = \frac{16}{23}, \text{ then what is the mean of}$$

a, b and c ?

[CDS 2019 (1)]

- (a) 1
(b) 2
(c) 1.33
(d) 2.33

105. What is the remainder when $(17^{29} + 19^{29})$ is divisible by 18?

[CDS 2019 (1)]

- (a) 6
(b) 2
(c) 1
(d) 0

106. What is the largest value of n such that 10^n divides the product:

$$2^5 \times 3^3 \times 4^8 \times 5^3 \times 6^7 \times 7^6 \times 8^{12} \times 9^9 \times 10^6 \times 15^{12} \times 20^{14} \times 22^{11} \times 25^{15} ?$$

[CDS 2019 (1)]

- (a) 65
(b) 55
(c) 50
(d) 45

107. How many pairs (A, B) are possible in the number 479865AB if the number is divisible by 9 and it is given that the last digit of the number is odd?

[CDS 2019 (1)]

- (a) 5
(b) 6
(c) 9
(d) 11

108. Consider the multiplication $999 \times abc = def132$ in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and respectively?

[CDS 2019 (1)]

- (a) 6,6,8,6,8,7
(b) 8,6,8,6,7,8
(c) 6,8,8,7,8,6
(d) 8,6,8,8,6,7

109. Which of the following statements is not true?

[CDS 2019 (1)]

- (a) The difference of two prime numbers, both greater than 2, is divisible by 2.
- (b) For two different integers m , n and a prime number p , if p divides the product $m \times n$, then p divides either m or n .
- (c) If a number is of the form $6n-1$ (n being a natural number), then it is a prime number.
- (d) There is only one set of three prime numbers such that there is a gap of 2 between two adjacent prime numbers.

110. The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the number is:

[CDS 2019 (1)]

- (a) 17
- (b) 29
- (c) 43
- (d) None of the above

111. What is the unit place digit in the expansion of 7^{73} ?

[CDS 2019 (1)]

- (a) 1
- (b) 3
- (c) 7
- (d) 9

112. Suppose n is a positive integer such that $(n^2 + 48)$ is a perfect square. What is the number of such n ?

[CDS 2019 (1)]

- (a) One
- (b) Two
- (c) Three
- (d) Four

113. Consider the following statements in respect of two integers p and q (both > 1) which are relatively prime:

- 1. Both p and q may be prime numbers
- 2. Both p and q may be composite numbers
- 3. One of p and q may be prime and the other composite

Which of the above statements are correct?

[CDS 2019 (1)]

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

114. Which is $0.\overline{53} + 0.5\overline{3}$ equal to?

[CDS 2019 (1)]

- (a) $1.\overline{068}$
- (b) $1.0\overline{68}$
- (c) $1.\overline{068}$
- (d) 1.068

115. Which one of the following is an irrational number?

[CDS 2019 (1)]

- (a) $\sqrt{59049}$
- (b) $\frac{231}{593}$
- (c) 0.45454545.....
- (d) 0.12112211122211112222.....

116. The number 3^{521} is divided by 8. What is the remainder?

[CDS 2019 (1)]

- (a) 1
- (b) 3
- (c) 7
- (d) 9

117. A prime number contains the digit X at unit's place. How many such digits of X are possible?

[CDS 2019 (1)]

- (a) 3
- (b) 4
- (c) 5
- (d) 6

118. The expression $5^{2n} - 2^{3n}$ has a factor:

[CDS 2019 (1)]

- (a) 3
- (b) 7
- (c) 17
- (d) None of the above

119. Consider two digit numbers which remain the same when the digits interchange their positions. What is the average of such digit numbers?

[CDS 2019 (1)]

- (a) 33
- (b) 44
- (c) 55
- (d) 66

120. If $\frac{36}{11} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, when x , y and z are

natural numbers, then what is $(x + y + z)$ equal to?

[CDS 2019 (2)]

- (a) 6
- (b) 7
- (c) 8
- (d) 9

121. Consider the following statements:

- 1. Unit digit in 17^{174} is 7:
- 2. Difference of the squares of any two odd numbers is always divisible by 8.
- 3. Adding 1 to the product of two consecutive odd numbers makes it a perfect square.

Which of the above statements are correct?

- (a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only
- 122.** When N is divided by 17, the quotient is equal to 182. The difference between the quotient and the remainder is 175. What is the value of N?
[CDS 2019 (2)]
(a) 2975 (b) 3094
(c) 3101 (d) 3269
- 123.** Consider the following statements:
1.If p is relatively prime to each of q and r, then p is relatively prime to the product qr.
2.If p divides the product qr and if p divides q, then p must divide r.
Which of the above statements is/are correct?
[CDS 2019 (2)]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 124.** Let x be the smallest positive integer such that when 14 divides x, the remainder is 7; and when 15 divides x, the remainder is 5. Which one of the following is correct?
[CDS 2019 (2)]
(a) $20 < x < 30$ (b) $30 < x < 40$
(c) $40 < x < 50$ (d) $x > 50$
- 125.** Consider the following statements:
1. $\sqrt{75}$ is a rational number
2. There exists at least a positive integer x such that $-\frac{4x}{5} < -\frac{7}{8}$.
3. $\frac{x-2}{x} < 1$ for all real values of x.
4. 4.232323..... can be expressed in the form $\frac{p}{q}$ where p and q are integer.
Which of the above statements are correct?
[CDS 2019 (2)]
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 2 and 4
- 126.** If 10^n divides $6^{23} \times 75^9 \times 105^2$, then what is the largest value of n?
[CDS 2019 (2)]
(a) 20 (b) 22
(c) 23 (d) 28
- 127.** What is the digit in the unit's place of the number represented by $3^{98} - 3^{89}$?
- (a) 3 (b) 6
(c) 7 (d) 9
- 128.** How many pairs of natural numbers are there such that the difference of their squares is 35?
[CDS 2019 (2)]
(a) 1 (b) 2
(c) 3 (d) 4
- 129.** Which one of the following is not correct?
[CDS 2019 (2)]
(a) 1 is neither prime nor composite
(b) 0 is neither positive nor negative
(c) If $p \times q$ is even, then p and q are always even
(d) $\sqrt{2}$ is an irrational number
- 130.** What is the sum of all integer values of n for which $n^2 + 19n + 92$ is a perfect square?
[CDS 2019 (2)]
(a) 21 (b) 19
(c) 0 (d) -19
- 131.** The number $2 \times 3 \times 5 \times 7 \times 11 + 1$ is:
[CDS 2020 (1)]
(a) A prime number
(b) Not a prime, but power of prime
(c) Not a power of a prime, but a composite even number
(d) Not a power of a prime, but a composite odd number
- 132.** If the sum of digit of a number $10^n - 1$, where n is natural number is equal to 3798, then what is the value of n?
[CDS 2020 (1)]
(a) 421 (b) 422
(c) 423 (d) 424
- 133.** If m is the number of prime numbers between 0 and 50; and n is the number of prime numbers between 50 and 100, then what is (m - n) equal to?
[CDS 2020 (1)]
(a) 4 (b) 5
(c) 6 (d) 7
- 134.** Which one of the following is the largest divisor of $3^x + 3^{x+1} + 3^{x+2}$, if x is any natural number?
[CDS 2020 (1)]
(a) 3 (b) 13
(c) 39 (d) 117

- 135.** A two digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two digit number is 8, then what is the number?

[CDS 2020 (1)]

- (a) 81 (b) 42
(c) 24 (d) 18

- 136.** If m and n are positive integers such that $m^n = 1331$, then what is the value of $(m-1)^{n-1}$?

[CDS 2020 (1)]

- (a) 1 (b) 100
(c) 121 (d) 125

- 137.** If the points P and Q represent real numbers $0.\overline{73}$ and $0.5\overline{6}$ on the number line, then what is the distance between P and Q ?

[CDS 2020 (1)]

- (a) $\frac{1}{6}$ (b) $\frac{1}{5}$
(c) $\frac{16}{45}$ (d) $\frac{11}{90}$

- 138.** The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order of the digits, is:

[CDS 2020 (1)]

- (a) Six (b) Five
(c) Four (d) Three

- 139.** Let XYZ be a 3-digit number. Let $S = XYZ + YZX + ZXY$. What of the following statements is/are correct?

1. S is always divisible by 3 and $(X+Y+Z)$
2. S is always divisible by 9
3. S is always divisible by 37

Select the correct answer using the code given below:

[CDS 2020 (1)]

- (a) 1 only (b) 2 only
(c) 1 and 2 (d) 1 and 3

- 140.** $x^3 + x^2 + 16$ is exactly divisible by x , where x is a positive integer. The number of all such possible values of x is:

[CDS 2020 (2)]

- (a) 3 (b) 4
(c) 5 (d) 6

- 141.** The number of (a, b, c) where, a, b, c are positive integers such that $abc = 30$, is:

[CDS 2020 (2)]

- (a) 30 (b) 27
(c) 9 (d) 8

- 142.** If the number 23P62971335 is divisible by the smallest odd composite number, then what is the value of P ?

[CDS 2020 (2)]

- (a) 4 (b) 5
(c) 6 (d) 7

- 143.** What is the remainder when the sum $1^5 + 2^5 + 3^5 + 4^5 + 5^6$ is divided by 4?

[CDS 2020 (2)]

- (a) 0 (b) 1
(c) 2 (d) 3

- 144.** What is the digit in the unit place of 3^{99} ?

[CDS 2020 (2)]

- (a) 1 (b) 3
(c) 7 (d) 9

- 145.** If 17^{2020} is divided by 18, then what is the remainder?

[CDS 2020 (2)]

- (a) 1 (b) 2
(c) 16 (d) 17

- 146.** The sum of all possible products taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$ is:

[CDS 2020 (2)]

- (a) 0 (b) -30
(c) -55 (d) 55

- 147.** Let $d(n)$ denote the number of positive divisors of a positive integer n . Which of the following are correct?

1. $d(5) = d(11)$
2. $d(5) \cdot d(11) = d(55)$
3. $d(5) + d(11) = d(16)$

Select the correct answer using the code given below:

[CDS 2020 (2)]

- (a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

- 148.** If $A_n = P_n + 1$, where P_n is the product of the first n prime numbers, then consider the following statements:

1. A_n is always a composite number
2. $A_n + 2$ is always an odd number
3. $A_n + 1$ is always an even number

Which of the above statements is/are correct?

[CDS 2020 (2)]

(a) 1 only
(c) 3 only

(b) 2 only
(d) 2 and 3 only

- 149.** The sum of the digits of a two digit number is 13 and the difference between the number and that formed by reversing the digits is 27. What is the product of the digits of the number?

[CDS 2020 (2)]

(a) 35
(c) 45

(b) 40
(d) 54

- 150.** If radius of a sphere is rational, then which of the following is/are correct?

1.Its surface area is rational
2.Its volume is rational

Select the correct answer using the code given below:

[CDS 2020 (2)]

(a) 1 only
(c) Both 1 and 2

(b) 2 only
(d) Neither 1 nor 2

- 151.** If the number 413283P759387 is divisible by 13, then what is the value of P?

[CDS 2021 (1)]

(a) 3
(c) 7

(b) 6
(d) 8

- 152.** What is the remainder when $2^{1000000}$ is divided by 7?

[CDS 2021 (1)]

(a) 1
(c) 4

(b) 2
(d) 6

- 153.** Consider the pairs of prime numbers (m,n) between 50 and 100 such that $m-n=6$. How many such pairs are there?

[CDS 2021 (1)]

(a) 2
(c) 4

(b) 3
(d) 5

- 154.** What is the unit digit in the expansion of 67^{32} ?

[CDS 2021 (1)]

(a) 1
(c) 7

(b) 3
(d) 9

- 155.** What are distinct prime factors of the number 26381?

[CDS 2021 (2)]

(a) 29, 17, 37
(c) 19, 37, 13

(b) 31, 17, 47
(d) 23, 31, 37

- 156.** What is the remainder after dividing the number 37^{1000} by 9?

[CDS 2021 (2)]

(a) 1
(c) 7

(b) 3
(d) 9

- 157.** If x is negative real number, then which of the following are not correct?

1. There is some natural number k such that $kx > 0$

2. $x^2 + x > 0$ always

3. $2x < x < -x$

4. x^2 is always a rational number

Select the correct answer using the code given below:

[CDS 2021 (2)]

(a) 1, 2 and 3
(c) 1, 3 and 4

(b) 1, 2 and 4
(d) 2, 3 and 4

- 158.** How many zeros are there in the product $150 \times 249 \times 348 \times \dots \times 501$?

[CDS 2021 (2)]

(a) 262
(c) 246

(b) 261
(d) 235

- 159.** A two digit number is such that the product of the digits is 8. If 63 is added to this number, the digits interchanged their places. What is the sum of the digit in the number?

[CDS 2021 (2)]

(a) 6
(c) 8

(b) 7
(d) 9

- 160.** Consider the following statements:

1. The sum of the cubes of three consecutive natural numbers is divisible by 9.

2. Every even power of every odd number (>1) when divided by 8 gives 1 as remainder.

Which of the above statements is/are correct?

[CDS 2022 (1)]

(a) 1 only
(c) Both 1 and 2

(b) 2 only
(d) Neither 1 nor 2

- 161.** What is the number of divisors of 1000 (excluding 1 and 1000)?

[CDS 2022 (1)]

(a) 12
(c) 14

(b) 13
(d) 16

- 162.** $27^5 + 3^{13}$ is divisible by:

[CDS 2022 (1)]

(a) 8
(c) 12

(b) 10
(d) 21

- 163.** Let $p = 2^{2n+2} + m$ and $q = 2^{4n} - m$ (where n is even natural number). What should be

the least value of m such that p as well as q is divisible by 5?

[CDS 2022 (1)]

- (a) -1 (b) 1
(c) 4 (d) 6

164. Let a, b, c, d be positive integers. If

$$\frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}} = \frac{17}{60}$$

then what is the product of a, b, c, d ?

[CDS 2022 (1)]

- (a) 24 (b) 51
(c) 68 (d) 102

165. What is the value of least value of n if $194480 + n = m^4$, where m and n are natural numbers?

[CDS 2022 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

166. $41^{43} + 43^{43}$ is divisible by

[CDS 2022 (1)]

- (a) 80 (b) 84
(c) 86 (d) 88

167. $46^1 + 46^2 + 46^3 + 46^4$ is divisible by:

[CDS 2022 (1)]

- (a) 7 (b) 9
(c) 11 (d) 17

168. There are two natural numbers x and y , where $x > y$. when x is divided by 6, it leaves the remainder 2 and; when y is divided by 6, it leaves the remainder 3. What is the remainder when $(x-y)$ is divided by 6?

[CDS 2022 (II)]

- (a) 1
(b) 3
(c) 5
(d) Remainder cannot be determined

169. If three times the greater of two numbers is divided by the smaller number, the quotient will be 6 and the remainder will be 6. If five times the smaller number is divided by the greater number, the quotient will be 2 and the remainder will be 3. What is the difference between the numbers?

[CDS 2022 (II)]

- (a) 8 (b) 9

- (c) 10 (d) 12

170. Consider the number $N = 12^6 \times 3^8 \times 5^3$. Which of the following statements is/are correct?

1. the number of odd factors of N is 60
 2. the number of even factors of N is 720
- Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

171. Consider the question and two statements given below:

A 2-digit number is added to the number formed by reversing the digits of the 2-digit number.

Question: What is the 2-digit number?

Statement-1: The sum is divisible by 9

Statement-2: The sum is divisible by 2.

Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

- (a) Statement-1 alone is sufficient to answer the question
(b) Statement-2 alone is sufficient to answer the question
(c) Both statement-1 and statement-2 are sufficient to answer the question
(d) Both statement-1 and statement-2 are not sufficient to answer the question

172. Let a, b, c and d be four positive integers such that $a + b + c + d = 200$. If $S = (-1)^a + (-1)^b + (-1)^c + (-1)^d$, then what is the number of possible values of S ?

[CDS - 2023 (1)]

- (a) one (b) two
(c) three (d) four

173. The number $97^{30} - 14^{30}$ is divisible by:

[CDS - 2023 (1)]

- (a) 37 but not 83 (b) 83 but not 37
(c) both 37 and 83 (d) neither 37 nor 83

174. Consider the following statements:

1. $\log_{10} 50$ is a rational number

2. $\log_{100} 10$ is an irrational number

Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

175. Consider the following statements:

1. $n^3 - n$ is divisible by 6
2. $n^5 - n$ is divisible by 5
3. $n^5 - 5n^3 + 4n$ is divisible by 120.

Which of the statements given above are correct?

[CDS - 2023 (1)]

- | | |
|------------------|------------------|
| (a) 1 and 2 only | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1, 2 and 3 |

176. What is the last digit of the sum $S = 9^{27} + 27^9$?

[CDS - 2023 (1)]

- | | |
|-------|-------|
| (a) 3 | (b) 6 |
| (c) 7 | (d) 9 |

177. Consider the following statements in respect of all factors of 360?

1. The number of factors is 24
2. The sum of all factors is 1170

Which of the above statements is/are correct?

[CDS - 2023 (1)]

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) both 1 and 2 | (d) neither 1 nor 2 |

178. Consider a 6-digit number of the form XYXYXY. The number is divisible by:

[CDS - 2023 (1)]

- | | |
|-----------------------|---------------------|
| (a) 3 and 7 only | (b) 7 and 13 only |
| (c) 3, 13 and 37 only | (d) 3, 7, 13 and 37 |

179. Let m and n be natural numbers. What is the minimum value of $(m + n)$ such that $33m + 22n$ is divisible by 121?

[CDS - 2023 (1)]

- | | |
|-------|--------|
| (a) 3 | (b) 4 |
| (c) 5 | (d) 10 |

180. If a , b and c are the sides of a triangle ABC, then $\sqrt{a} + \sqrt{b} - \sqrt{c}$ is always:

[CDS - 2023 (1)]

- | | |
|------------------|------------------|
| (a) Negative | (b) Positive |
| (c) non-negative | (d) non-positive |

181. When every even power of every odd integer (greater than 1) is divided by 8, what is the remainder?

[CDS - 2023 (1)]

- | |
|--|
| (a) 3 |
| (b) 2 |
| (c) 1 |
| (d) the remainder is not necessarily 1 |

182. Consider the following statements:

1. If n is a natural number, then the number $\frac{n(n^2 + 2)}{3}$ is also a natural number

2. If m is an odd integer, then the number $\frac{m^4 + 4m^2 + 11}{16}$ is an integer.

Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) both 1 and 2 | (d) neither 1 nor 2 |

183. It is given that 5 does not divide $n-1$, n and $n+1$, where n is a positive integer. Which one of the following is correct?

[CDS - 2023 (1)]

- | | |
|---------------------------|---------------------------|
| (a) 5 divide $(n^2 + 1)$ | (b) 5 divides $(n^2 - 1)$ |
| (c) 5 divides $(n^2 + n)$ | (d) 5 divides $(n^2 - n)$ |

184. What is the largest power of 10 that divides the product $29 \times 28 \times 27 \times \dots \times 2 \times 1$?

[CDS - 2023 (1)]

- | | |
|-------|-------|
| (a) 4 | (b) 5 |
| (c) 6 | (d) 7 |

185. What is the remainder when 65^{99} is divided by 11?

[CDS - 2023 (1)]

- | | |
|-------|--------|
| (a) 0 | (b) 5 |
| (c) 9 | (d) 10 |

186. Consider the following numbers:

1.437
2.797
3.1073

How many of the above numbers are prime?

[CDS - 2023 (1)]

- | | |
|---------------|--------------|
| (a) Only one | (b) Only two |
| (c) All three | (d) None |

For the next ten (4) items that follows:

Each item contains a question followed by two Statement. Answer each item using the following instructions:

[CDS - 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements

together, but cannot be answered by using either statement alone
(d) Choose this option if the question cannot be answered even by using both statements together.

- 187.** A number 277XY5 (where X, Y are digits) is divisible by 25.

Question: what is the value of X?

Statement I: The given number is divisible by 9

Statement II: $X > 5$.

Each item contains a question followed by two Statement. Answer each item using the following instructions:

- 188.** Question: Is $m > n$, where m, n are non-zero numbers?

Statement I: $\frac{m}{n} > 1$

Statement II: $m > 2n$

- 189.** x and y are consecutive odd integers.

Question: Can the value of $(x+y)$ be determined uniquely?

Statement I: $(x+y)^4 = 256$

Statement II: $(x+y)^3 < 16$

- 190.** Question: Is $p^2 + q^2 + q$ odd, where p, q are positive integers?

Statement I: $2p + q$ is odd

Statement II: $q - 2p$ is odd

Consider the following for the next three items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statement alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

[CDS-2023 (2)]

- 191.** Question: Is $m > n$ if m, n are real numbers?

Statement I:

$$M = (1-p)(p^2 + p + 1) \text{ and}$$

$$n = (p + 1)(p^2 - p + 1)$$

Statement II:

$$m = pn$$

- 192.** Question: What is the 3 digit number which is divisible by 10?

Statement I:

If the digits in hundred's place and ten's place of the number are interchanged, the resulting number is diminished by 180.

Statement II:

If the digit in hundred's place is halved and digit in ten's place and unit place of the number are interchanged, the resulting number is diminished by 336.

- 193.** Question: Is $(a-b+c) > (a+b-c)$, where a, b and c are real numbers?

Statement I:

b is negative

Statement II:

c is negative

- 194.** A 2-digit number is such that the sum of the number and the number obtained by reversing the order of the digits of the number is 55. Further, the difference of the given number and the number obtained by reversing the order of the digits of the number is 45. What is the product of the digits?

[CDS-2023 (2)]

(a) 5

(b) 2

(c) 1

(d) 0

Consider the following for the next four items that follow:

Each item contains a Question followed by two Statements. Answer each item using the following instructions:

Choose option

(a) If the Question can be answered by one of the Statements alone, but not by the other.

(b) If the Question can be answered by either Statement alone

(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.

(d) If the Question can be answered even by using both Statements together

- 195.** Let a, b, c and d be positive integers.

Question: Which one of a, b, c, d is closest to the product abcd?

Statement-I: $a > b > c$

Statement-II: c is not the smallest

[CDS-2024 (1)]

196. Let $mn = k$, where m and n are prime numbers and k is an even number.

Question: What is the value of $mn - n + 1$?

Statement-I: $m > n$

Statement-II: One of the numbers is 2

[CDS-2024 (1)]

197. Question: If p is a positive integer, then what is the remainder when p^n is divided by $p + 1$?

Statement-I: n is even

Statement-II: p is even

[CDS-2024 (1)]

198. Question: Is xy positive?

Statement-I: $x = \sqrt[3]{-0.19683}$

Statement-II: $y = \sqrt[3]{x}$

[CDS-2024 (1)]

199. What is the smallest natural number n such that $(n + 1) \times n \times (n - 1) \times (n - 2) \times \dots \times 3 \times 2 \times 1$ is divisible by 910?

[CDS-2024 (1)]

- (a) 91 (b) 90
(c) 13 (d) 12

200. The expression $555^{777} + 777^{555}$ is divisible by which of the following?

1. 2
2. 3
3. 37

Select the correct answer using the code given below:

[CDS-2024 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

201. Consider the following statements

1. If $(3m^3 + 2m^2 + 5m + n)/m$ is not an integer, where m and n are integers, then n is not divisible by m

2. $5(8^m) + 2^{3m}$ is divisible by 48 for all whole numbers m

Which of the statements given above is/are correct?

[CDS-2024 (1)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

202. What is the digit at the 100th place of number $(225)^{40}$?

[CDS-2024 (1)]

- (a) 6 (b) 5
(c) 4 (d) 2

203. If a, b, c, d are natural numbers, then how many possible remainders are there when $1^a + 2^b + 3^c + 4^d$ is divided by 10?

[CDS-2024 (1)]

- (a) 3 (b) 4
(c) 5 (d) 6

204. If n is a natural number, then what is the sum of all distinct remainders of $4^n + 6^n + 9^n + 11^n$ when divided by 10 for various values of n?

[CDS-2024 (1)]

- (a) 3 (b) 4
(c) 6 (d) 7

205. When the number (12345678910111213 ... 99100) is divided by 16, what will be the remainder?

[CDS-2024 (1)]

- (a) 15 (b) 12
(c) 4 (d) 3

ANSWER KEY

1.	c	2.	d	3.	a	4.	b	5.	b	6.	d	7.	b	8.	c	9.	d	10.	c
11.	d	12.	b	13.	d	14.	c	15.	c	16.	c	17.	c	18.	a	19.	a	20.	c
21.	b	22.	c	23.	d	24.	a	25.	c	26.	d	27.	a	28.	d	29.	d	30.	a
31.	b	32.	c	33.	b	34.	b	35.	c	36.	b	37.	a	38.	a	39.	a	40.	c
41.	c	42.	b	43.	d	44.	b	45.	a	46.	c	47.	d	48.	c	49.	a	50.	a
51.	c	52.	c	53.	d	54.	c,d	55.	a	56.	c	57.	c	58.	c	59.	c	60.	c
61.	d	62.	c	63.	d	64.	B	65.	b	66.	b	67.	d	68.	b	69.	b	70.	c
71.	a	72.	b	73.	b	74.	d	75.	d	76.	c	77.	a	78.	c	79.	a	80.	d
81.	b	82.	d	83.	c	84.	b	85.	c	86.	a	87.	b	88.	c	89.	b	90.	d
91.	c	92.	d	93.	c	94.	a	95.	d	96.	d	97.	b	98.	c	99.	c	100.	c
101.	b	102.	c	103.	b	104.	b	105.	d	106.	a	107.	a	108.	d	109.	c	110.	d
111.	c	112.	c	113.	d	114.	a	115.	d	116.	b	117.	c	118.	c	119.	c	120.	a
121.	c	122.	c	123.	a	124.	b	125.	d	126.	a	127.	b	128.	b	129.	c	130.	d
131.	a	132.	b	133.	b	134.	b	135.	a	136.	b	137.	a	138.	c	139.	d	140.	c
141.	b	142.	a	143.	b	144.	c	145.	a	146.	a	147.	b	148.	d	149.	b	150.	c
151.	d	152.	b	153.	d	154.	a	155.	d	156.	a	157.	b	158.	a	159.	d	160.	b
161.	c	162.	b	163.	b	164.	a	165.	a	166.	b	167.	d	168.	c	169.	b	170.	c
171.	d	172.	c	173.	c	174.	d	175.	d	176.	b	177.	c	178.	d	179.	b	180.	b
181.	c	182.	c	183.	a	184.	c	185.	d	186.	a	187.	c	188.	d	189.	c	190.	d
191.	d	192.	c	193.	c	194.	d	195.	c	196.	d	197.	a	198.	a	199.	d	200.	d
201.	a	202.	a	203.	c	204.	b	205.	b										

CAPF PYQ

1. If the product of n positive numbers is unity, then their sum is:
[CAPF 2017]
(a) A positive integer (b) Divisible by n
(c) Equal to $n + 1/n$ (d) Never less than n
2. Consider the following number $n = [(6374)^{1793} \times (625)^{317} \times (313)^{49}]$. Which one of the following is the digit at unit place of n .
[CAPF 2017]
(a) 0 (b) 1
(c) 2 (d) 5
3. What is the largest value four n (natural number) such that 6^n divides the product of the first 100 natural numbers.
[CAPF 2018]
(a) 18 (b) 33
(c) 44 (d) 48
4. Suppose $72 = m \times n$, where m and n are positive integers such that $1 < m < n$. How many possible values of m are there?
[CAPF 2018]
(a) 5 (b) 6
(c) 10 (d) 12
5. Suppose xyz are three positive integers such that $x \leq y \leq z$ and $xyz = 72$. Which one of the following values of S yields more than one solution to the equation $x+y+z=S$?
[CAPF 2018]
(a) 13 (b) 14
(c) 15 (d) 16
6. If the number $2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20}$ is divisible by 10^n . Then which one of the following is maximum value of n ?
[CAPF 2018]
(a) 78 (b) 85
(c) 89 (d) 98
7. Which one of the following is the remainder when 10^{20} is divided by 7?
[CAPF 2018]
(a) 1 (b) 2
(c) 4 (d) 6
8. On simplifications the product $(x_1 + y_1)(x_2 + y_2) \dots (x_{10} + y_{10})$, how many such terms are there which will have only single x and rest y 's?
[CAPF 2018]
(a) 2^{10} (b) 10
(c) 20 (d) 1
9. What is the natural number n for which $3^9 + 3^{12} + 3^{15} + 3^n$ is a perfect cube of an integer?
[CAPF 2019]
(a) 10 (b) 11
(c) 13 (d) 14
10. When the digits of two-digit numbers are reversed, the number increases by 27, the sum of such two-digit numbers is:
[CAPF 2019]
(a) 235 (b) 249
(c) 213 (d) 180
11. Which one of the following is the greatest number by which the product of three consecutive even numbers would be exactly divisible?
[CAPF 2020]
(a) 12 (b) 24
(c) 48 (d) 64
12. A is the smallest positive integer which when divided by 9 and 12 leaves remainder 8. B is the smallest positive integer which when divided by 9 and 12 leaves remainder 5. Which one of the following is the value of $A-B$?
[CAPF 2020]
(a) 3 (b) 2
(c) 1 (d) 0
13. Suppose the n th term of a series is $1 + \frac{n}{2} + \frac{n^2}{2}$. If there are 20 terms in the series, then the sum of the series is equal to:
[CAPF 2021]
(a) 1360 (b) 1450
(c) 1500 (d) 1560

ANSWER KEY

1.	d	2.	a	3.	d	4.	c	5.	b	6.	d	7.	b	8.	b	9.	d	10.	b
11.	b	12.	a	13.	d														